



Scheme/ Answer Key for Valuation

Scheme of evaluation (marks in brackets) and answers of problems/key

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.TECH S1 (SPECIAL IMPROVEMENT) EXAMINATION, JANUARY 2021 (2019 SCHEME)

Course Code: PHT 100

Course Name: ENGINEERING PHYSICS A

PART A

Answer all questions, each carries 3 marks.

			Marks
1		Definitions of free oscillation and damped oscillation ----1.5 marks each.	(3)
2		3 Definitions -----1 mark each	(3)
3		Condition for minimum intensity $2\mu t \cos r = n\lambda$ ---1 mark, Explanation ---2 marks.	(3)
4		Definition---2marks, condition---1 mark.	(3)
5		Definition of matter waves- 1mark, energy equations $E = h\nu$ & $E = mc^2$ ---0.5marks each , Obtain $\lambda = h/mv$ ---1 mark	(3)
6		Proper explanation--- 3 marks.	(3)
7		Faradays laws: 2 marks, Lenz's law: 1 mark	(3)
8		Statement----1 mark, Significance---2 marks	(3)
9		Explanation of critical field ---2marks Equation for variation of critical field with temperature-- 1mark.	(3)
10		Definition----1mark, Working Principle---2marks.	(3)

PART B

Answer one full question from each module, each question carries 14 marks

11.		a)Writing the differential equation ----- 2 marks Solution $x = A \sin (pt - \Theta)$ ----- 1 mark Finding differentials and substitution ----- 2 marks Equating coefficients ----- 2 marks , Finding amplitude A by squaring and adding -----2 marks; Finding phase difference Θ --- 1 mark. b)speed $v = \omega/k$ or $f\lambda$ Ans =15.63m/s----1 mark, amplitude =5 cm---1 mark, frequency=$\omega/2\pi=3.98$ Hz -- -- 1 mark, initial phase at origin = $\pi/2$ rad ---1 mark	(10)
			(4)



12	a)Figure with explanation-----2 marks. Obtain differential equation by equating forces----3 marks. General wave equation ---1 mark. Prove that $v = \sqrt{T/m}$ --- 1mark. Expression for frequency ----1.5 marks Statement three laws ---0.5 mark each. b)Eqn---$2/2l[\sqrt{T/m}]$----1 mark, substitution---2marks, Result with unit----frequency = 202.07 Hz----1 mark.	(10) (4)
13	a)Write expression for radius of n^{th} dark ring ---1 mark, Derivation of expressions for wavelength with liquid film $\lambda = \mu[d^2_{(n+k)} - d^2_n]/4kR$ -----2 marks Expressions for wavelength with air film $\lambda = [D^2_{(n+k)} - D^2_n]/4kR$-----1mark. Obtain expression for refractive index $\mu = \{D^2_{(n+k)} - D^2_n\} / \{d^2_{(n+k)} - d^2_n\}$ ----2 marks Explanation of experiment -----4 marks b)Equation---$d = \lambda/2\beta$-----1mark ,substitution with $d = 2r$ -----2marks Result with unit--- $4.91 \times 10^{-6}\text{m}$ -----1 mark	(10) (4)
14	a)Properly labelled diagram with explanation----2marks Obtaining path difference ----2marks Obtaining grating equation ----2marks Resolving power and dispersive power with expressions-----2 marks each b) Eqn---1 mark, substitution---1mark,finding $n(=4)$ ---1 mark , Result -----$N = 2.17 \times 10^5$ lines/m ---1 mark	(10) (4)
15	(a)Statement and explanation ----2marks ,Three forms of expression ----3marks Prove that the total energy of electron, $E = pc \geq 10\text{MeV}/20\text{MeV}$ ----4 marks, Explanation ----1mark (b)Equation $E = n^2 h^2/8mL^2$ ----1mark, $E_1 = 37.68 \text{ eV}$----2marks $E_2 = 4 E_1 = 150.72 \text{ eV}$,$E_3 = 9 E_1 = 339.12 \text{ eV}$ (reduce 1 mark if energy is in Joules)	(10) (4)
17	(a)Any 5 points for each as comparison----- 10 marks (b) $\chi = M / H$-----1 mark; Substitution and finding $\chi = 15.5$-----1 mark $\mu_r = 1 + \chi$ -----1 mark and , $\mu_r = 16.5$----- 1 mark.	(10) (4)



18	a)Maxwell's 4 equations----- 2 marks Derivation of the wave equation satisfied by electric field ---- 4 marks Writing wave equation satisfied by magnetic field ---- 2 marks Comparison with standard wave equation and finding the expression for velocity ----- 2 marks b)Poynting vector equation =Power/area ----1mark. Substitution and $S = \frac{3.8 \times 10^{26}}{4\pi \times (7 \times 10^8)^2} = 6 \times 10^7 W/m^2$ result with unit----3 marks.	(10) (4)
19	a)High temp superconductors----3 marks, Any 3 differences of Type I and Type II superconductors ---- 3 marks, graphs---1 mark each, examples--1 mark each. b)Any 4 applications -----4 marks	(10) (4)
20	a)Block diagram of optical fibre communication system---- 2 marks, Explanation of its functional blocks --- 3 marks. Any 3 advantages ----- 3 marks. b)Explain Sensors---2 marks, working of intensity modulator sensor---4 marks	(8) (6)

